

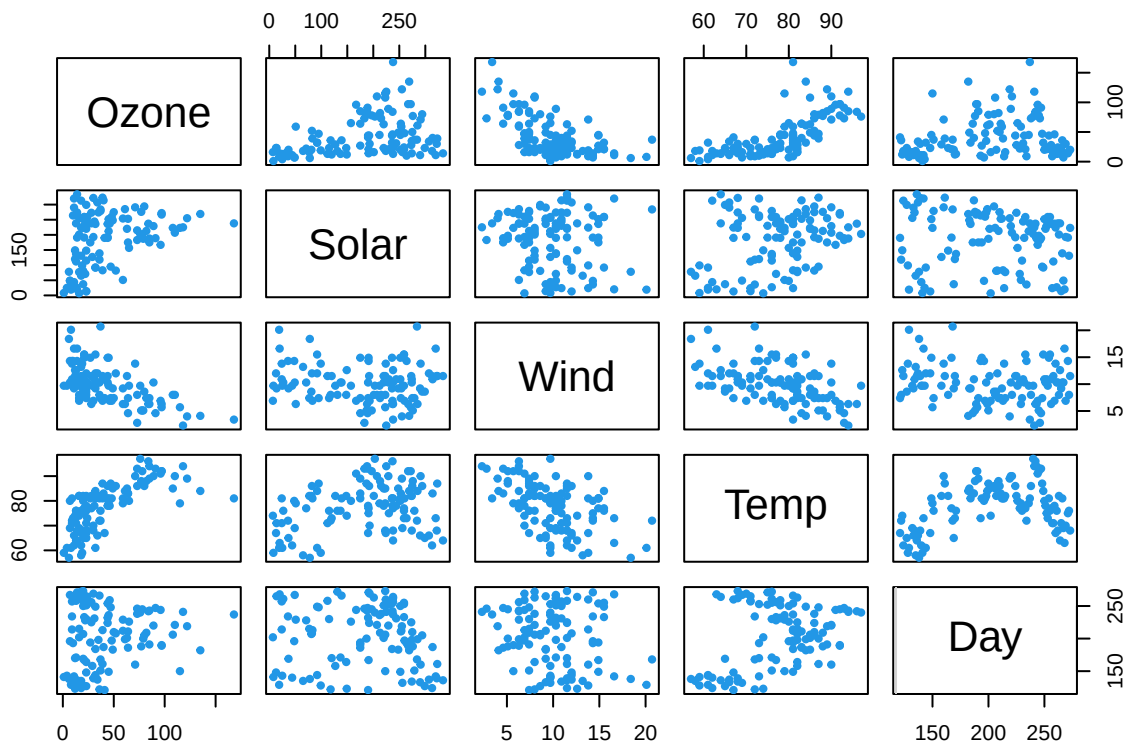
Ozone: an example of regression modelling

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Importing the data

```
ozone<-read.table("ozone.txt",header=TRUE)
plot(ozone,pch=20,col=4)
grid(lty=1)
```



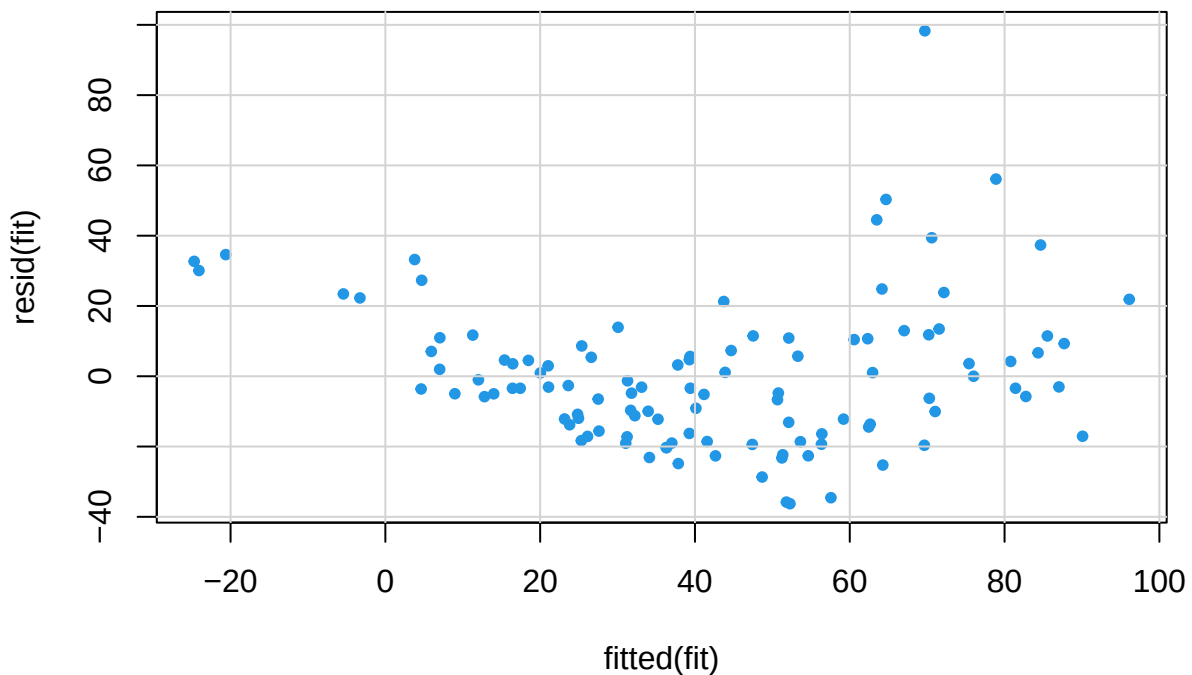
Fitting a linear model

```
fit <- lm(Ozone~Solar+Wind+Temp+Day, data=ozone)
summary(fit)

##
## Call:
## lm(formula = Ozone ~ Solar + Wind + Temp + Day, data = ozone)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -36.273 -14.126  -3.406   9.865  98.299
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -59.98323   23.00829  -2.607   0.0104 *
## Solar         0.05107    0.02358   2.166   0.0325 *
## Wind        -3.31806    0.64898  -5.113 1.42e-06 ***
## Temp         1.82684    0.27210   6.714 9.68e-10 ***
## Day         -0.08086    0.04816  -1.679   0.0961 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 21 on 106 degrees of freedom
## Multiple R-squared:  0.6161, Adjusted R-squared:  0.6016
## F-statistic: 42.53 on 4 and 106 DF,  p-value: < 2.2e-16
```

Residual analysis

```
plot(fitted(fit),resid(fit),pch=20,col=4)
grid(lty=1)
```



Confidence intervals for the estimates

```
Days <- data.frame(Solar=c(180,180), Wind=c(5,15), Temp=c(90,70), Day=c(274,274))
predict(fit, Days, interval="confidence")
```

```
##           fit           lwr           upr
## 1 74.880163 66.280450 83.47987
## 2  5.162824 -5.548428 15.87408
```

Prediction intervals

```
predict(fit, Days, interval="prediction")
```

```
##           fit           lwr           upr
## 1 74.880163 32.36081 117.3995
## 2  5.162824 -37.83335 48.1590
```

Transformation of the response

```
fit.log <- lm(log(Ozone)~Solar+Wind+Temp+Day,data=ozone)
summary(fit.log)
```

```
##
## Call:
## lm(formula = log(Ozone) ~ Solar + Wind + Temp + Day, data = ozone)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.1020 -0.2998 -0.0060  0.3012  1.2028
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.2099949  0.5579460  -0.376  0.707393
## Solar        0.0024105  0.0005717   4.216 5.24e-05 ***
## Wind        -0.0613767  0.0157377  -3.900 0.000169 ***
## Temp         0.0512613  0.0065984   7.769 5.35e-12 ***
## Day         -0.0009671  0.0011680  -0.828 0.409507
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.5093 on 106 degrees of freedom
## Multiple R-squared:  0.6666, Adjusted R-squared:  0.654
## F-statistic: 52.98 on 4 and 106 DF,  p-value: < 2.2e-16
```

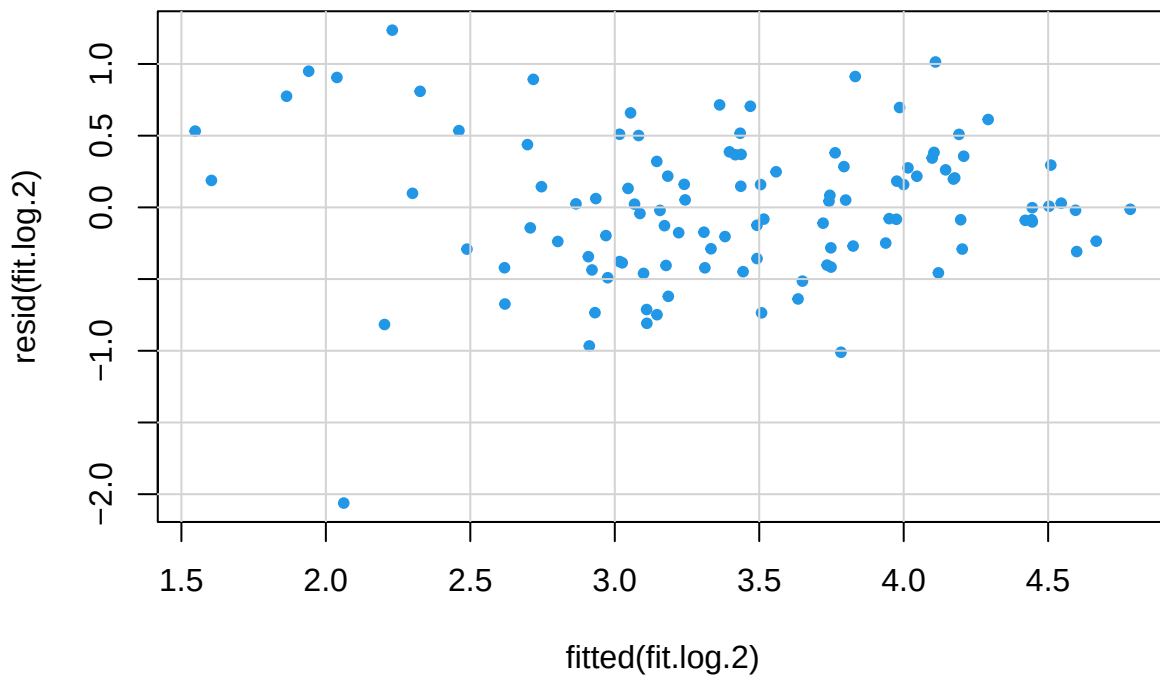
Model can be reduced

```
fit.log.2 <- lm(log(Ozone)~Solar+Wind+Temp,data=ozone)
summary(fit.log.2)
```

```
##
## Call:
## lm(formula = log(Ozone) ~ Solar + Wind + Temp, data = ozone)
##
```

```
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.06193 -0.29970 -0.00231  0.30756  1.23578
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.2621323  0.5535669  -0.474  0.636798
## Solar        0.0025152  0.0005567   4.518 1.62e-05 ***
## Wind        -0.0615625  0.0157130  -3.918 0.000158 ***
## Temp         0.0491711  0.0060875   8.077 1.07e-12 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.5086 on 107 degrees of freedom
## Multiple R-squared:  0.6644, Adjusted R-squared:  0.655
## F-statistic: 70.62 on 3 and 107 DF,  p-value: < 2.2e-16
```

```
plot(fitted(fit.log.2),resid(fit.log.2),pch=20,col=4)
grid(lty=1)
```



```
pred.log.2<-predict(fit.log.2, Days,interval="confidence")
pred.log.2
```

```
##      fit      lwr      upr
## 1 4.308188 4.126650 4.489726
## 2 2.709141 2.541013 2.877270
```

Estimates and confidence intervals on the original scale

```
pred<-exp(pred.log.2)
pred
```

```
##      fit      lwr      upr
## 1 74.30575 61.97001 89.09706
## 2 15.01637 12.69252 17.76570
```